1a)

$$\left(\begin{array}{ccc|c} 20'000 & 30'000 & 10'000 & 5'200'000 \\ 10'000 & 17'000 & 6'000 & 3'000'000 \\ 2'000 & 3'000 & 2'000 & 760'000 \end{array}\right) \begin{matrix} \\ -0.5 \\ -0.1 \end{matrix} \quad \left(\begin{array}{ccc|c} 20 & 30 & 10 & 5200 \\ 0 & 2 & 1 & 400 \\ 0 & 0 & 1 & 240 \end{array}\right)$$

$z = 240$

$2y + 1z = 400$

$2y + 240 = 400$

$2y = 160$

$y = 80$

$2x + 3y + 1z = 520$

$2x + 240 + 240 = 520$

$2x = 40$

$x = 20$

Hersteller A 20 Einheiten

B 80 –

– C 240 –

b)

$$A = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 5 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

c)

L

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 572 \\ 5 & 1 & 0 & 3300 \\ 1 & 0 & 1 & 836 \end{array}\right)$$

$x = 572$

$5x + y = 3300$

$2860 + y = 3760$

$y = 440$

$x + z = 836$

$572 + z = 836$

$z = 264$

R

$$\left(\begin{array}{ccc|c} 2 & 3 & 1 & 572 \\ 0 & 2 & 1 & 440 \\ 0 & 0 & 1 & 264 \end{array}\right)$$

$z = 264$

$2y + 1z = 440$

$2y + 264 = 440$

$2y = 176$

$y = 88$

$2x + 3y + 1z = 572$

$+264 + 264 = 572$

$2x = 44$

$x = 22$

Hersteller A 22 Einheiten

B 88 –

– C 264 –